

MWESIGYE EMMY

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SUMMARY

Software Development-focused computer scientist with strong hands-on experience in JavaScript, React.js, Node.js, Express, and databases, currently pursuing a Master of Science in Information Technology. Experienced in building full-stack applications, RESTful APIs, and scalable systems using modern development tools and cloud-native technologies. Adept at translating complex data into usable, user-friendly interfaces, with a strong interest in building impactful software for real-world problems. Known for adaptability, clean code practices, and collaborative teamwork.

EDUCATION

Master of Science in Information Technology

August 2025 - May 2027

Carnegie Mellon University, Africa

- Major: Software engineering
- Minor: Networking Technologies

Bachelor's Degree

April 2021 - June 2025

University of Rwanda, Kigali, Rwanda.

Information Security

PROFESSIONAL WORK EXPERIENCE

Internship

Feb 2021 - July 2021

Cisco Networking Academy – Networking & IT Essentials

- Attained CCNA certificates in Routing and Switching Networks.
- Attained ITE certificates in IT Essentials.

PROFESSIONAL SKILLS

- **Programming & Web development Skills:** Java, JavaScript, Python, C/C++, Data Structures & Algorithms, React.js, Express.js, Node JS, RESTful API design.
- **Frontend & UI:** React ecosystem, Responsive UI development, Familiarity with Tailwind CSS.
- **Database and Data:** MySQL, MongoDB, PostgreSQL, Basic data modeling & querying.
- **Cloud & DevOps:** Kubernetes, Docker (containerized development), OpenStack, Hadoop
- **Security:** Malware Analysis (RISA), Cybersecurity Fundamentals, Network Security
- **Networking:** Enterprise network design, router/switch configuration
- **Tools:** Git, Linux, CI/CD fundamentals, Linux environments
- **Technical Skills:** MS Excel, MS Word, MS PowerPoint
- **Languages:** Fluent in Kinyarwanda and English, and a little French
- **Interpersonal:** Strong communication skills, cultural competency, and adaptability in science

PROJECTS

Mobile Money Fraud Detection System

Dec 2021 - Jan 2022

- Developed a mobile money fraud detection Android system with a group of developers that captures fraudsters in real time.

Enterprise Cloud Infrastructure

Aug 2025 - Dec 2025

- Built an Infrastructure-as-a-Service cloud environment using Kubernetes, Docker, OpenStack, and Hadoop. Configured clusters, deployed workloads, and implemented multi-node virtualization and cloud networking.

Advanced Networking Labs

Aug 2025 - Dec 2025

- Designed and implemented scalable enterprise network topologies with routing protocols, VLANs, OSPF, NAT, and security configurations.

Malware Analysis Training (RISA)

June 2024 - June 2024

- Completed hands-on malware analysis labs, including static and dynamic analysis, behavioral analysis, and

defensive techniques.

CERTIFICATIONS

- ICDL Profile Certificate (2023)
- Certificate of CCNA Routing and Switching (2021)
- Certificate of Cisco IT Essentials (2021)
- Coursera usable security Certificate (2020)
- Introduction to ArcGIS Pro, ArcGIS Enterprise Fundamentals – ESRI Rwanda (2025)
- Malware Analysis Training - RISA (2024)